A SHORT SELF-CONTAINED ROUTE TO THE FIRST SUPPLEMENTARY LAW

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Abstract. We give a compact, fully elementary account of quadratic residues modulo an odd prime, leading to Euler's criterion and to the first supplementary law of quadratic reciprocity: -1 is a square modulo p if and only if $p \equiv 1 \pmod 4$. We close with some classical consequences.

1. Introduction

The first supplementary law of quadratic reciprocity states that -1 is a quadratic residue modulo an odd prime p exactly when $p \equiv 1 \pmod 4$. It is usually obtained either from Euler's criterion or from Wilson's theorem; see, e.g., [2, Ch. 5]. The purpose of this note is to collect, in one place and with complete proofs, the short chain of elementary facts leading to it, in a form suitable for a first course. Our main result is Theorem 2.8.

Throughout, p denotes an odd prime and congruences are modulo p unless stated otherwise. We use freely Fermat's little theorem: $x^{p-1} \equiv 1$ whenever $p \nmid x$. We write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

2. Quadratic residues and Euler's criterion

Definition 2.1. Let a be an integer with $p \nmid a$. We say that a is a *quadratic residue* modulo p if the congruence $x^2 \equiv a$ has a solution $x \in \mathbb{Z}$, and a *quadratic non-residue* otherwise. For an arbitrary integer a the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Lemma 2.2. *Let x, y be integers with $p \nmid x$. Then $x^2 \equiv y^2$ if and only if $y \equiv x$ or $y \equiv -x$.*

Proof. If $y \equiv \pm x$ then clearly $y^2 \equiv x^2$. Conversely, if $x^2 \equiv y^2$ then $p \mid x^2 - y^2 = (x-y)(x+y)$, and since p is prime, $p \mid x-y$ or $p \mid x+y$. □

Proposition 2.3. *The integers $1^2, 2^2, \dots, \left(\frac{p+1}{2}\right)^2$ are pairwise incongruent modulo p, and every quadratic residue modulo p is congruent to one of them. In particular, exactly $\frac{p-1}{2}$ of the integers $1, 2, \dots, p-1$ are quadratic residues modulo p.*

Proof. Let a be a quadratic residue, say $a \equiv x^2$. Then $p \nmid x$, so we may take $1 \le x \le p-1$; replacing x by $p-x$ if necessary (which does not change x^2 modulo p) we may assume $x \le \frac{p-1}{2}$. Now suppose that $x^2 \equiv y^2$ with $1 \le x < y \le \frac{p-1}{2}$. By Lemma 2.2, $p \mid y-x$ or $p \mid y+x$, which is impossible since $0 < y-x < p$ and $0 < x+y < p$. Hence the squares are pairwise incongruent, and the quadratic residues fall into exactly $\frac{p-1}{2}$ classes modulo p; each class meets $\{1, \dots, p-1\}$ in exactly one element. □

Corollary 2.4. *The sum of the quadratic residues modulo p lying in $\{1, 2, \dots, p-1\}$ is divisible by p.*

Proof. By Proposition 2.3 this sum is congruent to

$$\sum_{x=1}^{(p-1)/2} x^2 = \frac{1}{6} \cdot \frac{p-1}{2} \cdot \frac{p+1}{2} \cdot p = p \cdot \frac{(p-1)(p+1)}{24}.$$

Now $(p-1)(p+1)$ is a product of two consecutive even numbers, hence divisible by 8; and among the three consecutive integers $p-1, p, p+1$ one is divisible by 3, and it is not p. Thus $24 \mid (p-1)(p+1)$ and the sum is divisible by p. □

Lemma 2.5 (Lagrange)**.** *A nonzero polynomial $f \in \mathbb{F}_p[X]$ of degree d has at most d roots in $\mathbb{F}_p$.*

Proof. Induction on d; the case $d = 0$ is clear since f is then a nonzero constant. If $d \ge 1$ and $f(c) = 0$, division by $X - c$ gives $f = (X-c)g$ with $g \ne 0$ of degree $d-1$. If $c' \ne c$ is another root then $(c'-c)g(c') = 0$ and, $\mathbb{F}_p$ being a field, $g(c') = 0$. So f has at most $1 + (d-1) = d$ roots. □

Lemma 2.6. *If $p \nmid a$, then $a^{(p+1)/2} \equiv 1$ or $a^{(p+1)/2} \equiv -1$.*

Proof. Put $y = a^{(p-1)/2}$. By Fermat's little theorem $y^2 = a^{p-1} \equiv 1 = 1^2$, so Lemma 2.2 with $x = 1$ gives $y \equiv \pm 1$. □

Theorem 2.7 (Euler's criterion)**.** *For every integer a,*

$$a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

Proof. If $p \mid a$, both sides are divisible by p, because $(p-1)/2 \ge 1$. Let $p \nmid a$. If a is a quadratic residue, $a \equiv x^2$ with $p \nmid x$, and $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$. By Proposition 2.3 there are $\frac{p-1}{2}$ distinct classes of quadratic residues in $\mathbb{F}_p$, and we have just seen that each of them is a root of $X^{(p-1)/2} - 1$. By Lemma 2.5 this polynomial has no further roots. Hence if a is a non-residue then $a^{(p-1)/2} \not\equiv 1$, and Lemma 2.6 forces $a^{(p-1)/2} \equiv -1$. □

Theorem 2.8. *Let p be an odd prime. Then -1 is a quadratic residue modulo p if and only if $p \equiv 1 \pmod 4$.*

Proof. Apply Theorem 2.7 with $a = -1$: $(-1)^{(p-1)/2} \equiv \left(\frac{-1}{p}\right)$. Both sides lie in $\{1, -1\}$, since $p \nmid -1$, and $1 \not\equiv -1$ because $p > 2$; so $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ as integers. As p is odd, $(p-1)/2$ is even exactly when $p \equiv 1 \pmod 4$, which proves the claim. □

3. Consequences

We recall two classical theorems.

Theorem 3.1 (Wilson; see [1, Thm. 80])**.** $(p-1)! \equiv -1 \pmod p$.

Proposition 3.2. *If* $p \equiv 1 \pmod 4$ *and* $x = \left(\frac{p-1}{2}\right)!$, *then* $x^2 \equiv -1 \pmod p$.

Proof. Pair each $k \in \{1, \dots, \frac{p-1}{2}\}$ with $p - k \equiv -k$. By Theorem 3.1,

$$-1 \equiv (p-1)! = \prod_{k=1}^{(p-1)/2} k(p-k) \equiv \prod_{k=1}^{(p-1)/2} (-k^2) = (-1)^{(p+1)/2} x^2.$$

Since $p \equiv 1 \pmod 4$, the exponent $(p+1)/2$ is even, and $x^2 \equiv -1$. □

Proposition 3.2 makes the "if" direction of Theorem 2.8 explicit. The following theorem of Fermat and Euler gives yet another witness.

Theorem 3.3 (Fermat–Euler; see [1, Ch. XX])**.** *Let* $n > 1$ *be an integer all of whose prime factors are congruent to* 1 *modulo* 4. *Then there is exactly one pair of integers* (a, b) *with* $0 < a < b$ *and* $n = a^2 + b^2$.

Indeed, if $p = a^2 + b^2$ is prime then $p \nmid b$, and $(ab^{-1})^2 \equiv -1$.

Corollary 3.4. *Let* $p \equiv 3 \pmod 4$ *and let* x, y *be integers with* $p \mid x^2 + y^2$. *Then* $p \mid x$ *and* $p \mid y$.

Proof. If $p \nmid y$, let z be an inverse of y modulo p. Then $(xz)^2 \equiv -y^2 z^2 \equiv -1$, so -1 is a quadratic residue, contradicting Theorem 2.8. Hence $p \mid y$, then $p \mid x^2$, and so $p \mid x$. □

References

[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.

[2] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer, 1990.

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